

Problem 69. Let n be a positive integer. Show that there exists an integer m such that

$$2020m^2 + 20202019m + 2019$$

is divisible by 2^n .

Solution. We shall show more generally that

$$am^2 + bm + c \equiv 0 \pmod{2^n}$$

has a solution for all positive integers n whenever b is odd and a or c is even.

For $n = 1$, take $m = 0$ if c is even and $m = 1$ if c is odd.

Now suppose the claim is true for n . If c is even, then by assumption, the congruence

$$2at^2 + bt + \frac{c}{2} \equiv 0 \pmod{2^n}$$

has a solution in t . Letting $m = 2t$, we get

$$am^2 + bm + c = 2\left(2at^2 + bt + \frac{c}{2}\right) \equiv 0 \pmod{2^{n+1}}.$$

If c is odd, then a is even and thus $a + b + c$ is even; hence by assumption, the congruence

$$2at^2 + (2a + b)t + \frac{a + b + c}{2} \equiv 0 \pmod{2^n}$$

has a solution in t . Letting $m = 2t + 1$, we get

$$am^2 + bm + c = 2\left(2at^2 + (2a + b)t + \frac{a + b + c}{2}\right) \equiv 0 \pmod{2^{n+1}}.$$

This completes the proof of the claim by induction. □

Problem 70. Find the largest positive integer n such that there exist n real polynomials where the sum of any two has no real roots but the sum of any three does.

Solution. The answer is $n = 3$.

When $n = 3$, we can take the following three constant polynomials $f, g, h \equiv -1, -2, 3$ and they clearly satisfy the problem conditions.

Now assume that $n = 4$ and let the polynomials be $f_1(x), f_2(x), f_3(x), f_4(x)$. Note that for any i, j , we must have either

$$f_i(x) + f_j(x) > 0 \quad \text{or} \quad f_i(x) + f_j(x) < 0 \quad \text{for all } x$$

as otherwise it must have a real root. If there exist indices i, j, k such that

$$f_i(0) + f_j(0), \quad f_i(0) + f_k(0), \quad f_j(0) + f_k(0)$$

all have the same sign, say positive, then for all x ,

$$f_i(x) + f_j(x) > 0, \quad f_i(x) + f_k(x) > 0, \quad f_j(x) + f_k(x) > 0,$$

and so

$$f_i(x) + f_j(x) + f_k(x) > 0,$$

which is a contradiction as the sum $f_i(x) + f_j(x) + f_k(x)$ would have no real roots.

We shall show that such a triple of indices must exist. Without loss of generality let

$$|f_1(0)| \geq |f_i(0)| \quad \text{for } i = 2, 3, 4$$

and that $f_1(0) > 0$. Then $f_1(0) + f_i(0) > 0$ for all i . If there exist i, j chosen from $2, 3, 4$ such that $f_i(0) + f_j(0) > 0$, then $\{1, i, j\}$ is such a triple. If not, then $\{2, 3, 4\}$ is such a triple. Thus n cannot be four. Therefore the only answer is $n = 3$. □

Problem 71. Choose positive integers $x_1, x_2, \dots$ satisfying

$$1 = \frac{x_1}{1^2} > \frac{x_2}{2^2} > \frac{x_3}{3^2} > \frac{x_4}{4^2} > \dots$$

and let α denote the largest real number satisfying

$$\frac{x_n}{n^2} \geq \alpha \quad \text{for all positive integers } n.$$

What are the possible values of α over all possible choices of the sequence $\{x_n\}$?

Solution. The answer is $0 \leq \alpha \leq \frac{1}{2}$.

Obviously $\alpha \geq 0$. In one direction, we show the following claim.

Claim (Greedy bound): For all integers n , we have

$$\frac{x_n}{n^2} \leq \frac{1}{2} + \frac{1}{2n}.$$

We prove this by induction on n . For $n = 1$ it is given. For the inductive step we have

$$\begin{aligned} x_n &< n^2 \cdot \frac{x_{n-1}}{(n-1)^2} \\ &\leq n^2 \cdot \left(\frac{1}{2} + \frac{1}{2(n-1)} \right) \\ &= \frac{n^3}{2(n-1)} \\ &= \frac{1}{2} \left(n^2 + n + 1 + \frac{1}{n-1} \right) \\ &= \frac{1}{2} n(n+1) + \frac{1}{2} \left(1 + \frac{1}{n-1} \right) \\ &\leq \frac{1}{2} n(n+1) + 1. \end{aligned}$$

Since x_n is an integer, we have $x_n \leq \frac{1}{2} n(n+1)$, and this implies the claim.

We now give a construction:

- for $\alpha = \frac{1}{2}$ we take $x_n = \frac{1}{2} n(n+1)$;
- for $\alpha = 0$ we take $x_n = 1$;
- for $0 < \alpha < \frac{1}{2}$, let N be large enough that

$$\lceil \alpha n^2 + n \rceil < \frac{1}{2} n(n+1) \quad \text{for all } n \geq N.$$

Then the following sequence works:

$$x_n = \begin{cases} \lceil \alpha n^2 + n \rceil, & n \geq N, \\ \frac{1}{2} n(n+1), & n < N. \end{cases}$$

Indeed we certainly have

$$\frac{x_n}{n^2} = \frac{\alpha n^2 + n + O(1)}{n^2} \rightarrow \alpha \quad \text{as } n \rightarrow \infty.$$

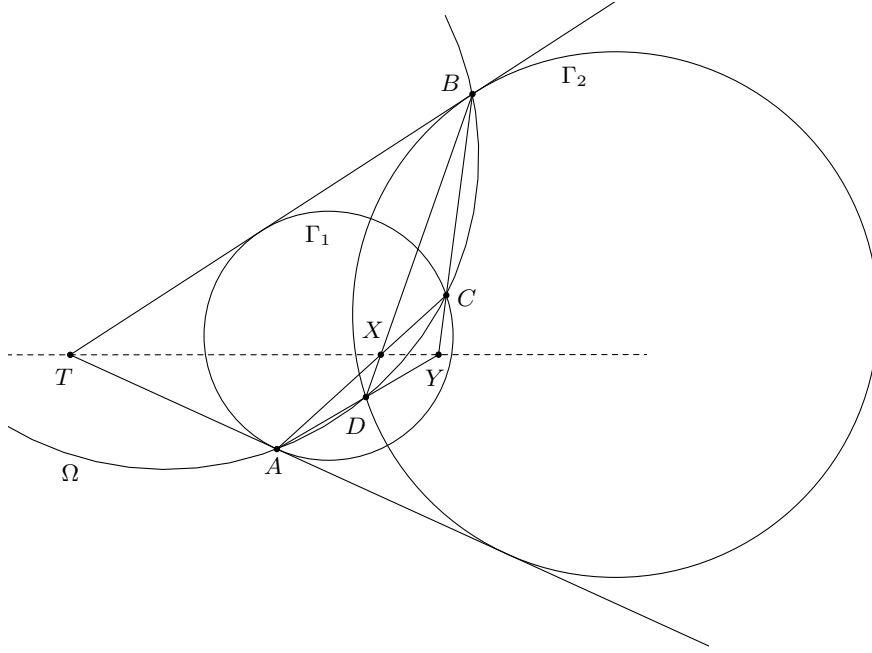
Mainly, we argue that $\frac{x_n}{n^2}$ is strictly decreasing. We need only check this for $n \geq N$. In fact, we have

$$\frac{x_n}{n^2} \geq \frac{\alpha n^2 + n}{n^2} > \frac{(\alpha(n+1)^2 + (n+1)) + 1}{(n+1)^2} > \frac{x_{n+1}}{(n+1)^2}$$

where the middle inequality is true since it rearranges to $\frac{1}{n} > \frac{n+2}{(n+1)^2}$. □

Problem 72. Two circles Γ_1 and Γ_2 have common external tangents ℓ_1 and ℓ_2 meeting at T . Suppose ℓ_1 touches Γ_1 at A and ℓ_2 touches Γ_2 at B . A circle through A and B intersects Γ_1 again at C and Γ_2 again at D , such that quadrilateral $ABCD$ is convex. Suppose lines AC and BD meet at point X , while lines AD and BC meet at point Y . Show that T, X, Y are collinear.

Solution 1.



Since $\triangle YAC \sim \triangle YBD$, we have

$$\frac{d(Y, AC)}{d(Y, BD)} = \frac{AC}{BD}.$$

Moreover, if we denote by r_1 and r_2 the radii of Γ_1 and Γ_2 , then

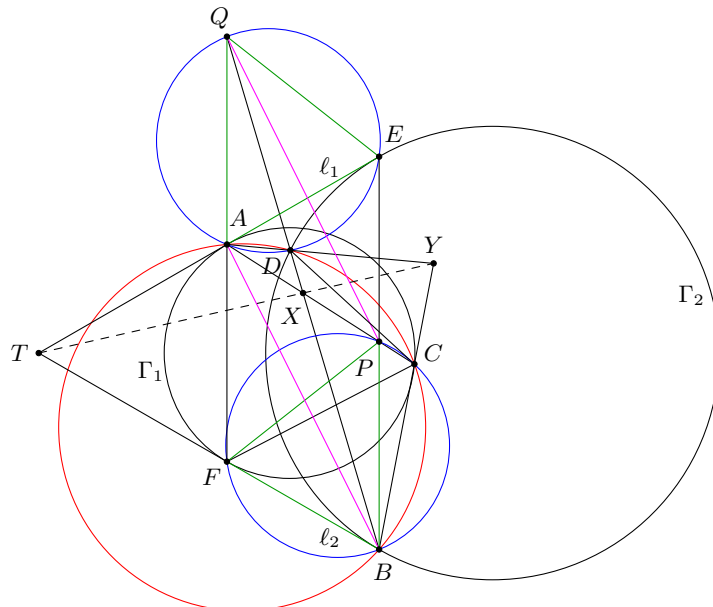
$$\frac{d(T, AC)}{d(T, BD)} = \frac{TA \sin \angle(AC, \ell_1)}{TB \sin \angle(BD, \ell_2)} = \frac{2r_1 \sin \angle(AC, \ell_1)}{2r_2 \sin \angle(BD, \ell_2)} = \frac{AC}{BD}$$

the last step by the sine formula.

This solves the problem up to configuration issues: we claim that Y and T both lie inside $\angle AXB = \angle CXD$. Without loss of generality assume that $TA < TB$.

- The former is since Y lies outside segments BC and AD , since we assumed $ABCD$ was convex.
- For the latter, we note that X lies inside both Γ_1 and Γ_2 in fact on the radical axis of the two circles (since X was an interior point of both chords AC and BD). In particular, X is contained inside $\angle ATB$, and moreover $\angle ATB < 90^\circ$, and this is enough to imply the result. $\square$

Solution 2.



Let ℓ_1 touch Γ_2 at E , ℓ_2 touch Γ_1 at F . Let AC meet BE at P , BD meet AF at Q . We claim that all of T, X, Y lie on the radical axis ℓ of (ADE) and (BCF) , hence proving that they are collinear.

Firstly, as TAE and TFB are tangents to Γ_1 and Γ_2 , we have $TA \times TE = TF \times TB$. So T lies on ℓ .

Secondly, as A, B, C, D are concyclic, we have $YA \times YD = YB \times YC$. So Y lies on ℓ .

Next, note that $AFBE$ is an isosceles trapezoid. Therefore, we have $\angle FBP = \angle TFA = \angle FCP$. This implies P lies on (BCF) . Similarly, Q lies on (ADE) . Since $\angle FBP = \angle EAQ$, $\angle BFP = 180^\circ - \angle BCP = 180^\circ - \angle BCA = 180^\circ - \angle BDA = \angle ADQ = \angle AEQ$ and $BF = AE$, we have $\triangle BFP \cong \triangle AEQ$. Therefore, $BP = AQ$ and $ABPQ$ is a parallelogram. It follows that $\angle XCD = \angle XBA = \angle XQP$. Thus, C, P, D, Q are concyclic, and $XC \times XP = XD \times XQ$. This shows X lies on ℓ and we are done.

□